Nageeb Zaman

(Md Nageeb Bin Zaman)

mzama14@lsu.edu — (+1) 225-409-0994 — Baton Rouge, LA
in/nageebzaman — Google Scholar — ORCID — U.S. Permanent Resident

Summary

Computational physicist and Ph.D. candidate (expected May 2027) with 7 years of experience developing, modifying, and running physics-based simulation codes in Fortran, C/C++, and Python on high-performance computing (HPC) systems. Experience spans multiphysics radiation-hydrodynamics, Monte Carlo radiative transfer, and density functional theory (DFT) electronic-structure modeling, plus GPU/CPU performance-portability work at Los Alamos National Laboratory. Builds end-to-end simulation and data-analysis pipelines and validates models against measured data.

Education

Ph.D. Candidate, Physics, Louisiana State University, Baton Rouge, LA (GPA 3.97/4.00) Expected May 2027
M.S., Physics, Central Michigan University, Mount Pleasant, MI 2021
M.S. and B.S., Physics, University of Dhaka, Dhaka, Bangladesh 2016, 2014

Technical Skills

Programming	Python, C/C++, Fortran, Bash, Java
Modeling Methods	Radiation hydrodynamics, Monte Carlo radiative transfer, DFT / electronic structure, numerical methods, parameter studies
Simulation Codes	MESA, STELLA, SuperLite, AresPK, AthenaPK, FLO-SIC, NRLMOL
HPC	Linux clusters, Slurm, MPI-parallel codes, CPU/GPU performance and scaling studies (NVIDIA A100); familiar with Kokkos, OpenMP
Data & Visualization	NumPy, SciPy, Pandas, Matplotlib, Astropy, MATLAB, Mathematica, Gnuplot, VESTA
Tools	Git/GitHub, VS Code, Vim, LaTeX

Research Experience

Graduate Research Assistant 2021–Present
Louisiana State University, Supernovae and Mergers Group, Baton Rouge, LA

- Built a multi-code simulation pipeline coupling binary stellar-evolution (MESA), 1D radiation-hydrodynamics (STELLA), and Monte Carlo radiative-transfer (SuperLite) codes to predict the time-dependent light output and spectra of stellar explosions.
- Designed and ran a grid of 30 circumstellar-interaction models (wind-like, compact-shell, and detached-shell structures, plus hydrogen-rich composition variants), mapping how density structure, radial extent, and composition shift peak luminosity, timescale, temperature, and spectral features (first-author paper under review, *ApJ*).
- Found that only massive detached shells sustain both superluminous brightness and hot, blue emission, matching the global properties of fast superluminous supernovae; showed that luminosity, duration, temperature, and spectra must be fit jointly, and traced remaining spectral mismatches to specific model limitations (LTE treatment, 1D geometry).
- Compared synthetic spectra with observed supernova spectra using a spectral-classification tool (Duperfit); currently fitting observed light curves to constrain ejecta mass and explosion energy for a new modeling study.
- Developed Python pipelines (NumPy, SciPy, Pandas, Astropy) to process hundreds of GB of simulation output, extract physical parameters, and produce publication-quality figures.
- Ran large simulation campaigns totaling 300,000+ core-hours on Slurm-managed HPC clusters, using MPI-parallel codes; modified, compiled, and debugged Fortran and C/C++ scientific software.
- Performed Monte Carlo radiative-transfer spectral analysis of shell-collision models, identifying the physical origin of time-varying spectral features (*ApJ*, 2025).

Graduate Intern, Co-Design Summer School Summer 2023
CCS-7 (Applied Computer Science), Los Alamos National Laboratory, Los Alamos, NM

- One of a four-member team that built AresPK (originally Ares), a performance-portable multiphysics hydrodynamics code for thermonuclear (Type Ia) supernova explosions, extending the Parthenon adaptive-mesh-refinement (AMR) framework with Kokkos to run on both CPUs and GPUs.
- Implemented the nuclear-statistical-equilibrium (NSE) burning solver that computes energy release from thermonuclear

burning; the team also added a monopole self-gravity solver and the first GPU implementation of the Helmholtz equation of state (via LANL's Singularity-EOS), which matched the reference implementation to machine precision.
- Ran strong- and weak-scaling studies on LANL's Chicoma supercomputer (AMD EPYC CPUs, NVIDIA A100 GPUs): perfect strong scaling up to 8 GPUs, one A100 GPU on average 34% faster than 64 CPU cores, and a largest run of a 1024^3 mesh on 512 MPI ranks (32,768 CPU cores). Presented at SC23 (poster) and the LANL summer-school exit talk.

Graduate Research Assistant 2019–2021
Central Michigan University, Computational Condensed Matter Physics Group, Mount Pleasant, MI

- Modified and extended the Fortran-based FLO-SIC density functional theory (DFT) code to speed up self-interaction-corrected electronic-structure calculations of molecules.
- Implemented and benchmarked a Frozen Potential approximation that bypasses repeated potential evaluations, cutting force-evaluation runtime by 72–86% on test molecules (HCN, C_2H_4, CH_3F) while reproducing the exact forces' behavior near the optimal configuration; analyzed results in Python and visualized structures with VESTA. (M.S. thesis; talk at APS March Meeting 2021.)

Graduate Teaching Assistant LSU 2021–2023; CMU 2020–2021
Taught undergraduate physics laboratories; mentored students in instrumentation, data analysis, and technical reporting. Received CMU Best Teaching Assistant Award (2020).

Publications

- **Zaman, N.**, & Chatzopoulos, E. "Photometric and Spectroscopic Diagnostics of Hydrogen-poor Circumstellar Interaction in Stripped-envelope Transients." *The Astrophysical Journal* (2026), under review.
- Huynh, T. N., Chatzopoulos, E., & **Zaman, N.** "Rotational Dynamics in Pulsational Pair-Instability Supernovae: Implications for Mass-Loss and Transient Events." *The Astrophysical Journal* (2025).

Selected Presentations

- *Ares – Simulating Type Ia Supernovae on Heterogeneous HPC Architectures.* Poster, SC23 International Conference for High Performance Computing, Denver, CO (2023).
- *Diagnostics of Hydrogen-poor Circumstellar Interaction in Stripped-envelope Transients.* Talks at AAS Winter Meeting, New Orleans (2024) and Rise Time 2024, Purdue University; posters at CraigFest 2024, UT Austin, and Transients from Space 2025, STScI/Johns Hopkins University.
- *Accelerating Fermi-Löwdin Orbital Descriptor Optimization for Self-Interaction-Free DFT.* Talk, APS March Meeting (2021).

Awards

Roussel Family Graduate Student Award, Dept. of Physics & Astronomy, Louisiana State University 2024
Best Teaching Assistant Award, Dept. of Physics, Central Michigan University 2020

Leadership

Webmaster, LSU Physics Graduate Student Organization 2022–2023